REVISION No. - . **CHANGE NO. - .**

MARINE PROPELLER PURCHASE SPECIFICATION 48” x 40” (Inland River Tenders with IRESS upgrades)

IBCT-PL P-245-0146

ORGINATED BY J. Happe . DATE 03/15 .

ORIGINATOR’S COMMAND Surface Force Logistics Command (SFLC) .

DEPARTMENT / DIVISION IBCT-PL .

1.0. SCOPE

This specification sets forth the requirements for the manufacturing, testing, inspection and packaging of marine propellers for the Inland River Tenders of 48” diameter. The number and type of propellers to be purchased in accordance with this specification will be indicated on the Contract Deliverables List.

2.0. APPLICABLE DOCUMENTS

2.1. DOCUMENT AVAILABILITY - To obtain the referenced documents see section 6.1.

2.2. GOVERNMENT DOCUMENTS - The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the documents listed here by number and date shall be those in the current Department of Defense Single Stock Point of Specifications and Standards (DODSSP) and supplements thereto.

2.2.1. SPECIFICATIONS, MILITARY

a. MIL-DTL-2845E (SH) Dated 18 August 1999, Preservation, Packaging, Packing and Storage of Propulsion Systems, Boat & Ship; Main Shafting, Propellers, Bearings, Gauges, Special Tools, & Associated Repair Parts.

2.3. OTHER GOVERNMENT DOCUMENTS - The following documents form a part of this specification to the extent specified herein. The documents listed shall be by title and effective date, unless otherwise documented.

2.3.1. DRAWINGS

a. USCG Drawing No. 75-WAGL-4400-003, Rev.L, dated 03/2015

2.3.2. SPECIFICATIONS

a. COAST GUARD SPECIFICATION, D-000-0100, (latest revision), Bar Coding for USCG Surface Forces Logistics Center Material.

2.4. COMMERCIAL DOCUMENTS - The following documents form a part of this specification to the extent specified herein. The documents shall be the version listed by number and date from the referenced industry organization. Commercial documents may also be listed in the Department of Defense Single Stock Point of Specifications and Standards (DODSSP).

2.4.1. Society of Automotive Engineers (SAE)

a. SAE-J-755-80 Dated May 2015, Marine Propeller-Shaft Ends & Hubs.

b. SAE-J-448A - Dated February 2023, Surface Texture Standard.

2.4.2. American Bureau of Shipping (ABS)

a. ABS Guidance Manual for Bronze & Stainless Steel Propeller Castings Dated 1984.

2.4.3. International Organization for Standardization

a. ISO 484/2-1981 (E)- Shipbuilding - Ship screw propellers - Manufacturing tolerances - Part 2: Propellers of diameter between 0.80 m and 2.50 m inclusive.

2.4.4. American National Standards Institute (ANSI)/ National Conference of Standards Laboratories (NCSL).

a. ISO/IEC 17025, General Requirements for the competence of testing and calibration laboratories.

2.4.5. American Society For Quality (ASQ).

a. ANSI/ASQ Q9003-1994, Dated 199, Model for Quality Assurance Standards in final inspection and test. .

2.5. SPECIFICATION ORDER OF PRECEDENCE - In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3.0. REQUIREMENTS

3.1. GENERAL - Each propeller shall be classified to the National Stock Number (NSN), and the distinguishing characteristics listed in section 3.2. Manufactured propellers shall meet all requirements of this specification.

3.2. DISTINGUISHING CHARACTERISTICS - Each propeller is distinguished as detailed in Table 1 of this specification by NSN, Blade Orientation, Boat/Cutter Application, Diameter, Pitch, Rake, Skew, Number of Blades, Rated Rotational Speed (RPM), Nominal Hub Bore, Nominal Hub Length, Mean Width Ratio (MWR), and Developed Area Ratio (DAR). All propellers shall have an elliptical contour, vertical genetrix, true helical face, and segmental section.

3.2.1 DIAMETER - Diameter is the distance across a circle generated by the most extreme point of the blades. Propeller diameter shall be as listed in table 1.

3.2.2 PITCH - Pitch is the distance the generating helix forming the pressure face of the propeller advances in one rotation. Pitch, determined by measuring the pressure face of the propeller as specified in 3.8 , shall be as listed in table 1.

3.2.3 BLADE ORIENTATION - Blade orientation shall be determined in accordance with the “right hand rule”: Right hand rotation produces thrust directed along the thumb of the right hand when the propeller is rotated in the direction of the fingertip’s, left hand is the opposite.

3.2.4 DEVELOPED AREA RATIO - Developed area ratio (DAR) is the ratio of the developed blade area (the true area of the pressure face surface of all of the blades) divided by the area of a circle with the same diameter as the propeller. The DAR listed in table 1 is the minimum allowed, if specified.

3.2.5 MEAN WIDTH RATIO - Mean width ratio (MWR) is the ratio of the developed area of one blade divided by the length of the blade outside the hub and the diameter. The MWR listed in table 1 is the minimum allowed, if specified.

3.2.6 SKEW – Skew is determined by running a line approximately radially, connecting the line of maximum thickness, (or midline, for segmental propellers) of each section of a blade. The skew angle is the angle subtended along the circumference of the diameter by the midline from hub to blade tip. Propellers shall have either no skew or approximately 15 degrees of blade skew such that the trailing edge of the propeller is essentially straight within the 70% radius from the hub as listed in table 1.

3.2.7 BLADE THICKNESS - Blade thickness is determined by Blade Thickness Fraction, (the ratio between the maximum section blade chord and the diameter), or by the actual maximum section thickness at a specified section. Propellers shall have a blade thickness fraction of 0.036 to 0.051.

3.2.8 NOMINAL HUB BORE - Nominal hub bore is the diameter of the bore at the largest point of the taper. Nominal hub bore is listed in table 1.

3.2.9 RATED ROTATIONAL SPEED - Rated rotational speed is specified for determining residual unbalance in accordance with 3.7.

3.2.10 HUB LENGTH AND SPECIAL HUB CHARACTERISTICS – Certain propeller hubs have additional features for fitting sealing glands or other components. These features include but are not limited to tapped holes for screws, bored recesses, specific hub lengths, and other modifications to the hub. Special hub characteristics for a given propeller will be listed in the notes to table 1.

3.3. MATERIAL
All materials shall be ABS type 4 as specified herein and as listed in the ABS Guidance Manual section 1, part 1.2 (materials). Classification or certification with ABS is not required.

a. Ultimate Tensile 85,000 psi
b. Yield 35,000 psi
c. Elongation in 50 mm (2 in.) 15%

NOTE: Differing materials that may be listed on any of the drawings referenced in section 2.3.1 of this specification are superseded by this requirement. **All propellers shall be cast using ABS type 4 Nickel Aluminum Bronze.**

3.4. SURFACE FINISH – Surfaces of all propeller blades and hubs shall be polished to 125 micro inches or better in accordance with SAE-J-448A.

3.5. VISUAL INSPECTION - All propellers shall be free of flaws in accordance with the criteria of Table 1.1 and Figure 1.1 of the ABS Guidance Manual. **Note: Dye penetrant examination is not required.**

3.6. BORES, KEYWAYS, HUB DIMENSIONS & DETAILS - All bores, keyways, and hubs shall meet the dimensional requirements of SAE J755-80, table 3 for hubs without sleeves, including surface finish.

a. The nominal hub bore diameter shall be as specified for each propeller in the deliverable list.

b. The hub shall be the full length of the taper as a minimum in accordance with table 3 or table 4 of reference 2.4.1.a, unless otherwise noted in table 1.

c. Each propeller shall be marked in the hub area with the diameter and pitch in that order, separated by an “X”.

3.7. BALANCING - Each propeller shall be single plane balanced as stated herein. All balancing shall be accomplished by means of balancing equipment, which requires rotation of the work piece. The residual unbalance in each plane of correction shall not exceed the value determined by calculating the following formula: $\mathbf{U = 4000W/N^2}$ where:

U = Maximum allowable residual unbalance in oz.-inches
W = Weight of rotating part in pounds
N = Maximum operating speed (RPM) of propeller

Prior to performing any balance tests, the Contractor shall demonstrate the sensitivity of the balance machine using a weight determined from the following formula: **Ws = 0.5(U)/r** where:

Ws = Sensitivity Weight
U = Propeller residual unbalance tolerance
r = Radial location of the weight

The addition of this weight shall be easily detected and discernible to the balance machine operator.

3.7.1. The mandrel or concentric rings used to support the propeller during balancing shall not exceed a total indicated run-out (TIR) of .001 inches.

3.7.2. Material removed to effect balancing shall be taken from the suction face of the heavy blade. Care shall be taken to keep a fair surface.

a. The area from which metal is taken shall not come within 10 percent of the chord length to the leading or trailing edges.

3.7.3. Using W=319lbs and N=525, U is calculated to be 4.63 oz-in.

3.8. TOLERANCES - All tolerances shall be in accordance with drawing 75-WAGL-4400-003, sheet 1 general notes tolerances. In the absence of any tolerances in USCG Drawing, then ISO 484/2, Class I shall be followed.

a. METHODS FOR MEASURING PITCH - Pitch may be measured by a computer controlled coordinate measurement machine or a "Pitchometer" that can be shown to be calibrated to and to produce equivalent accuracy to the methods of Section 4 of ISO 484/2 for Class I propellers in addition to the methods of Section 4.

b. MEASUREMENT LOCATIONS - Measurement of a section near the hub shall not be required where specified by ISO 484/2.

3.9. TESTS & INSPECTIONS

a. The listed tests shall be performed on each ladle poured in accordance with the ABS Guidance Manual section 1, part 1.2.2 & 1.2.3, except that ABS classification or certification or witnessing by an ABS Surveyor is not required.

1. Tensile, Yield, & Elongation tests.

2. Material Chemical Analysis.

b. The Contractor shall perform a dimensional and visual inspection on each propeller to assure compliance with this specification and its references.

3.10. WORKMANSHIP - All propellers submitted to the Coast Guard for acceptance shall conform to the form, fit, finish, and material requirements specified herein.

3.11. PROPELLER STAMPING - Each propeller shall have the diameter, pitch, rotation, and serial number clearly stamped on the outside diameter of the aft end of the hub and clear of the fillets. Stamping shall be neat and legible, 1/4” to 1/2” high.

3.12. NOTIFICATIONS & INSPECTIONS - The Contractor shall be responsible for notifying the Contracting Officer at least seven (7) working days prior to inspections or monitored testing required herein.

4.0. QUALITY ASSURANCE (Q.A.)

4.1. GENERAL - The Contractor shall provide and maintain an inspection system which shall ensure each item offered to the Government for acceptance conforms with contract requirements. The system shall be documented and meet the requirements of ANSI/ASQ-Q9003-1994.

4.2. RESPONSIBILITY FOR INSPECTION - The Contractor is responsible for the performance of all inspection requirements as specified herein. The Contractor can use his own or any other facilities for the performance of the inspections and tests. The government reserves the right to perform any of the inspections and tests set forth in this specification at any time.

4.3. RESPONSIBILITY FOR COMPLIANCE - All items shall meet the provisions of Sections 3 & 5. The inspections set forth in this specification shall become part of the Contractor’s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the Contractor of the responsibility of assuring that all products or supplies submitted to the government for acceptance comply with all requirements of the contract. Sampling for quality conformance does not authorize submission of known defective material (either indicated or actual) nor does it commit the government to acceptance of defective material.

4.4. NONCOMPLIANCE - Shall mean any single failure of any individual test or inspection.

a. Any flaws detected in material or construction methods shall be resolved to the satisfaction of the Coast Guard before completing any further work.

4.5. INSPECTIONS/TESTS - The inspections and tests required herein are the minimum necessary and are not intended to replace any controls, examinations or tests normally employed by the Contractor to assure the quality of the product. The following are required and may be monitored by a Government Quality Assurance Representative.

4.5.1. Dimensional inspection in accordance with sections 3.6 and 3.8.

4.5.2. Pitch measurement inspection in accordance with section 3.8.

4.5.3. Surface finish in accordance with section 3.4.

4.5.4. Visual inspection in accordance with section 3.5.

4.5.5. Balancing as required in section 3.7.

4.5.6. Tensile, Yield, and Elongation tests in accordance with section 3.9.a.1.

4.5.7. Material chemical analysis in accordance with section 3.9.a.2.

4.5.8. Preservation, packaging, packing and marking in accordance with section 5.0.

4.6. TEST AND INSPECTION REPORTS - One copy of any required inspection and test report shall be sent to the Contracting Officer for approval prior to QA inspection.

a. A copy of any test or inspection report shall be turned over to the Coast Guard Quality Assurance Representative upon his arrival at the Contractor's plant.

4.7. CALIBRATION SYSTEM - The Contractor as well as any subcontractors shall be required to maintain a test equipment calibration program in compliance with ANSI/NCSL Z540-1. The program shall be documented and traceable to the National Institute of Standards (NIST).

a. The Contractor shall maintain a copy of this system along with all calibration records for the duration of any contract referencing this specification, and make them available for inspection on demand.

4.8. PACKAGING INSPECTION - The Contracting Officer shall be notified as in section **3.12** prior to the preservation, packaging, and marking of the completed units.

5.0. PRESERVATION & PACKAGING, PACKING, & MARKING

5.1. PRESERVATION

a. All propellers shall be cleaned in accordance with MIL-DTL-2845E section 3.7.1.

b. Propellers 30" in diameter and under shall be preserved in accordance with MIL-DTL-2845E section 3.7.10.

c. Propellers over 30" in diameter shall have strippable compound preservative applied to each propeller in accordance with MIL-DTL-2845E section 3.14.1.3.1.

5.2. PACKAGING

a. Propellers 30” in diameter and under shall be individually packed in accordance with MIL-DTL-2845E, section 3.14.2.2.1(b)(1).

b. Propellers over 30” in diameter shall be individually packed in accordance with MIL-DTL-2845E section 3.14.2.2.2.

5.3. MARKING - Each preserved, packed, and packaged item shall be stenciled in the upper left-hand side of the two longest sides, (two opposite sides if square) of each box. The lettering shall be at least 1 inch high, in black print lettering on a highly contrasting **painted** background.

5.3.1 This marking shall consist of the STOCK NUMBER, ITEM NAME, PART NUMBER, SERIAL NUMBER, CONTRACT NUMBER, SHIPPING WEIGHT (in lbs.,) condition code ie. CONDITION A, unit of issue with quantity (1) and the words COAST GUARD ELC MATERIAL.

5.3.2 Marking layout shall be as outlined below.

STOCK NUMBER:	*As indicated on contract*
ITEM NAME:	**PROPELLER,** (*specify diameter and pitch*)
PART NUMBER:	*As indicated on contract*
SERIAL NUMBER:	*To be provided by the Contracting Officer*
CONTRACT NUMBER:	*As indicated on contract*
DELIVERY ORDER NUMBER:	*As indicated on contract*
SHIPPING WEIGHT:	*As determined by Contractor*
CONDITION CODE:	**CONDITION A**
QUANTITY & UNIT OF ISSUE:	**1- EA**
COAST GUARD ELC MATERIAL	

5.4. BAR CODING - Bar code labels shall be supplied on all CLIN’s except for those that require data submissions only.

5.4.1. All bar coded labels shall be in accordance with ISO/IEC-16388-2007, Code 39 Symbology.

a. The actual labels shall be Type “V”, Grade “B”, Style “2” Composition “a” with plastic laminate.

b. Each label shall contain encoded data for the Stock Number.

c. The marking required in 5.3 can be applied separately or as part of the Bar Code Label.

6.0. NOTES

6.1. GOVERNMENT & COMMERCIAL DOCUMENT SOURCES

6.1.1. GOVERNMENT DOCUMENTS

a. Department of Defense Single Stock Point (DODSSP)
700 Robbins Avenue
Building #4, Section D
Philadelphia, PA 19111-5094
215-697-2664 FAX 215-697-9398

b. DOD & Navy Publications & Forms (Other Than DODSSP)
Naval Publication and Forms Center
5801 Tabor Ave.
Philadelphia, PA 19120
215-697-2000

6.1.2. COMMERCIAL DOCUMENTS

a. Society of Automotive Engineers, INC. (SAE)
400 Commonwealth Drive
Warrendale, PA 15096
724-776-4841 FAX 724-776-0790
www.sae.org

b. American Bureau of Shipping (ABS)
ABS Plaza
16855 Northchase Drive
Houston, TX 77060
281-877-6000, FAX 281-877-6001
www.eagle.org

c. International Organization for Standardization (ISO)
American National Standards Institute
1430 Broadway,
New York, NY 10018
212-642-4900

d. American Society for Quality (ASQ)
600 North Plankinton Avenue
Milwaukee, Wisconsin 53203
414-272-8575 FAX 414-272-1734

e. American National Standards Institute (ANSI)
25 West 43rd Street
(Between 5th and 6th Avenues) 4th Floor,
New York, NY 10036
212-642-4900 FAX 212-398-0023
www.ansi.org

6.1.3. OTHER GOVERNMENT DOCUMENTS

a. USCG Surface Forces Logistics Center
Mail Stop 26, Bldg. 31 Lower Level
2401 Hawkins Point Road
Baltimore, MD 21226-5000
Attn. IBCTPL Supply

Table 1

NSN	Blade Orientation	Boat/ Cutter Application	Diameter In Inches	Pitch In Inches	Rake In Degrees	Skew In Degrees	No. of Blades	RPM	Nominal Hub Bore In Inches	Nominal Hub Length In Inches	MWR	DAR
2010-01-534-9095	LH	65 WLR/ 75 WLIC	48.000	40.000	0	0	4	525	See Note 1	10.750	0.36	0.710
2010-01-534-9089	RH											